

## JSXGraph and AI: Making Dynamic Mathematics Understandable for Intelligent Assistants

### 1 What Is JSXGraph?

JSXGraph is an open-source JavaScript library for interactive mathematical visualizations in the browser, including 2D and 3D geometry, function graphs, and curves.

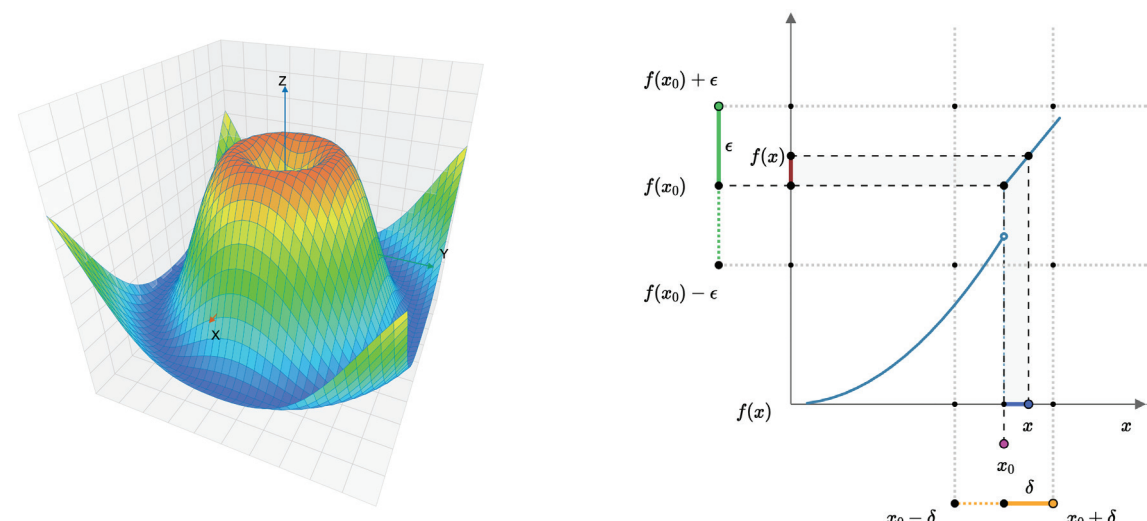


Fig. 1 3D and 2D function graphs

Objects can depend on one another: for example, a line depends on two points and a function graph can depend on a slider. When users move an object, JSXGraph automatically updates all dependent objects.



Fig. 2 Line depends on two points

### 2 JSXGraph Is Code-Based

JSXGraph constructions are written in JavaScript, making them well suited for AI-supported workflows.

- ▶ AI can generate and modify JSXGraph code.
- ▶ The code is compact and pattern-based.
- ▶ Many examples and tutorials are available online.

As a result, language models can often create code that appears correct at first glance.

### 3 The Main Problem

Static correctness is not enough. An AI-generated construction may look correct initially but fail during interaction.

For example, a triangle may appear right-angled, but its right-angle vertex is not constrained to lie on the circle with the opposite side as its diameter. When a point is moved, the right angle disappears.

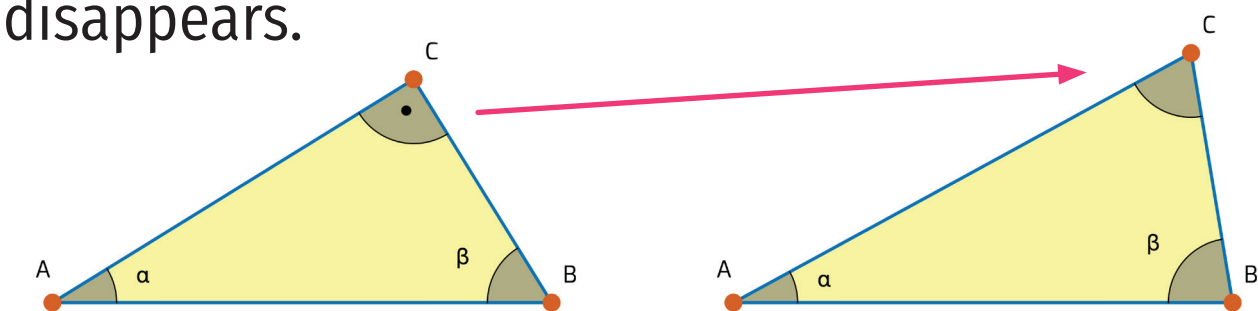


Fig. 3 Visual correctness does not guarantee dynamic validity.

The missing element is the dependency structure that preserves mathematical properties dynamically.

### 4 Current Situation

The typical JSXGraph AI workflow is as follows:

1. An AI system generates JSXGraph code.
2. The browser executes this JSXGraph code and renders the construction.
3. Internally, JSXGraph stores object relationships as a dependency graph of ancestors and descendants.
4. The AI system usually has access only to the source code and visible result – not to the construction's internal state.

As a result, the AI system cannot reliably determine whether a construction is dynamically valid.

### 5 Why This Is Challenging

AI assistants can read JSXGraph examples online and reuse coding patterns.

However, they cannot fully inspect or interpret a live construction.

At runtime, the AI has little information about:

- ▶ object dependencies,
- ▶ correctly encoded geometric relationships,
- ▶ changes caused by moving an object,
- ▶ and the construction's stability.

Additional structured information is therefore required – the central focus of this project.

### 6 Structural Runtime Data

The project proposes structured data exports (“dumps”) that provide AI systems with real-time information about a JSXGraph construction.

#### Code Dump

Provides the current JessieCode (JSXGraph's own construction language), allowing AI systems to inspect the board definition and compare revisions.

#### Dependency Information

Represents relationships between objects, including ancestors and descendants. It lets the AI system distinguish visual alignment from genuine dynamic and geometric dependencies.

#### Interaction Log

Records user interactions, especially object movements, and captures how the construction responds. This provides evidence about its stability and correctness.

### 7 Adding Structural Data

Dependency information could be published alongside JSXGraph example code. Example collections, tutorials, and user pages could include a machine-readable dump of ancestor-descendant relationships. This gives AI systems access not only to code, but also to a construction's intended structure and dynamic behavior.

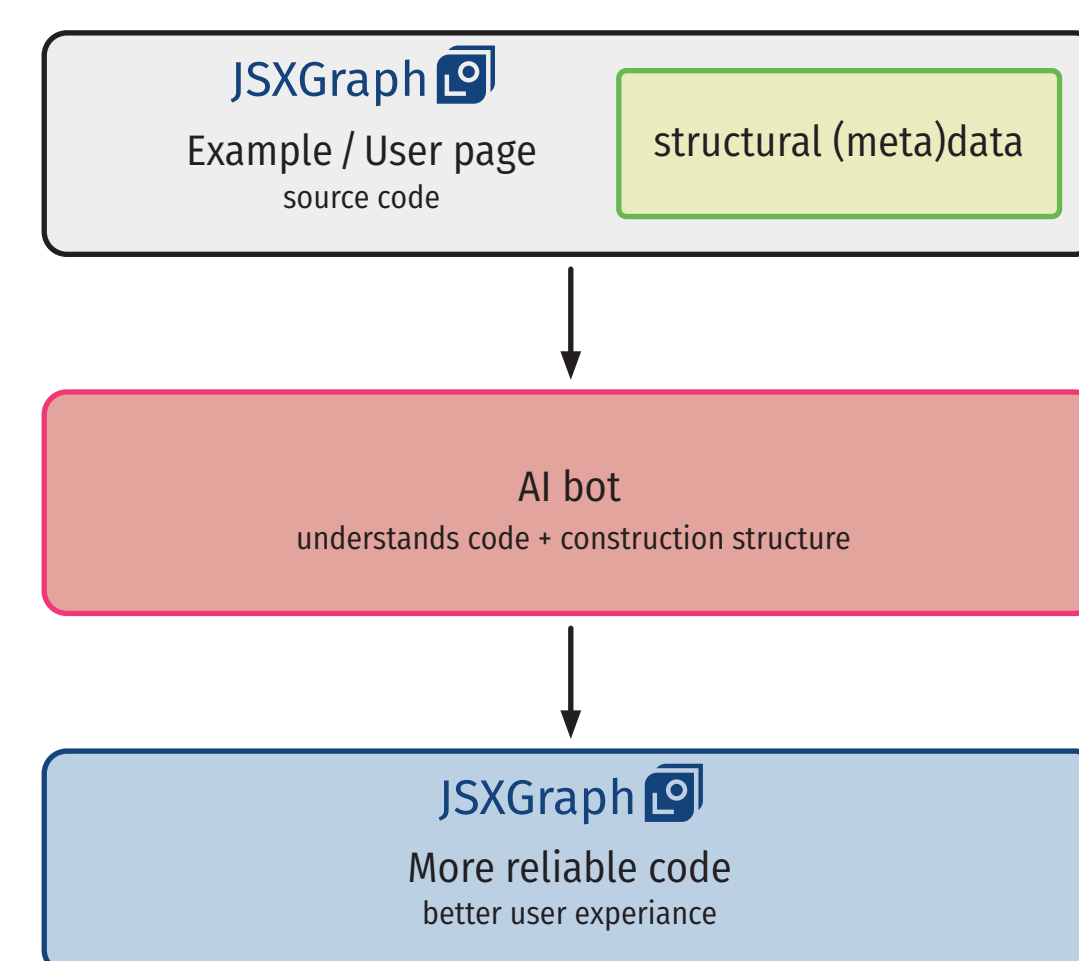


Fig. 4 From code-only AI to structure-aware AI-generated constructions

As a result, AI systems could learn essential dependencies and generate more reliable code before a construction is executed.

### 8 Project Goal

The goal is to give AI assistants access to the information needed to understand dynamic JSXGraph constructions – not only their code or visual appearance.

By combining code, dependency, and movement data, an AI system can:

- ▶ analyze whether dependencies are correctly implemented,
- ▶ identify unstable or incomplete constructions,
- ▶ explain geometric relationships,
- ▶ learn from structurally annotated examples,
- ▶ and generate more reliable JSXGraph code.

### 9 Key Message

For AI-generated dynamic mathematical constructions, correct initial output is not sufficient.

An AI system needs access to a construction's runtime structure and behavior. Exposing code, dependencies, and user interactions can turn AI systems into more capable assistants for dynamic mathematical constructions.